SQL Practice Problems

Scenario: Hogwarts

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Hogwarts Database Schema

Assume the following relational schema for Hogwarts School of Witchcraft and Wizardry:

```
Students(
    student_id    INT PRIMARY KEY,
    name          VARCHAR(100),
    house_id      INT,
    year          INT,          -- 1 to 7
    blood_status  VARCHAR(20)   -- 'Muggle-born', 'Half-blood', 'Pure-blood'
);

Houses(
    house_id   INT PRIMARY KEY,
    house_name VARCHAR(50),    -- Gryffindor, Slytherin, etc.
    head_id    INT             -- references Professors
);

Professors(
    professor_id INT PRIMARY KEY,
    name         VARCHAR(100),
    subject      VARCHAR(100)
);

Courses(
    course_id   INT PRIMARY KEY,
    course_name VARCHAR(100),
    professor_id INT
);

Enrollments(
    student_id INT,
    course_id  INT,
    grade      VARCHAR(2),    -- O, E, A, P, D, T
    PRIMARY KEY(student_id, course_id)
);

QuidditchMatches(
    match_id   INT PRIMARY KEY,
    house_home INT,
    house_away INT,
    goals_home INT,
    goals_away INT,
    match_date DATE
);
```

Foreign keys may be assumed even if not explicitly declared.

Part 1 – JOIN + GROUP BY + HAVING

Q1. Average Grades per Course (Only Popular Courses)

For each course with at least 5 enrolled students, show:

- course_name
- number of students
- average grade (converted to score):
  O=100, E=85, A=70, P=55, D=40, T=0

Q2. Most Popular Course in Each House

For each house, find the course with the most students from that house.

Q3. Houses That Win More Quidditch Matches Than They Lose

Q4. Students per House and Blood Status (At Least 3 Students)

Part 2 – Window Functions

Q5. Ranking Students by O+E Grades Within Each House

Q6. Top 2 Courses by Enrollment (Using Window Function)

Q7. Running Total of Quidditch Goals Per House

Q8. Percentile Rank of Students by Course Load